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Sub-THz vector load pull tuner

Abstract

Wideband waveguide slot-based bi-directional couplers are combined with waveguide load pull tuners for true vector load pull at sub-THz frequencies. Coupling is constant and Directivity is above average up to 170 GHz and can be extended to 330 GHz and are controlled by the shape, size, and configuration of the slots between the main tuner waveguide and the adjacent coupler waveguide. A calibration method allows full characterization of the coupler-tuner assembly.

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Background/Summary

PRIORITY CLAIM

(1) Not Applicable

CROSS-REFERENCE TO RELATED ARTICLES

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BACKGROUND OF THE INVENTION

(3) This invention relates to sub-Terahertz (THz) testing of transistors (device under test: DUT). The electrical signals injected into the input of the DUT and extracted from the output can be sampled and measured using signal sampling devices (directional couplers, see ref. 1 and 5) and processed by appropriate signal vector analyzers (see ref. 7).

DESCRIPTION OF PRIOR ART

(4) A typical test setup allowing sampling electrical signals at the input and output of a DUT in linear and nonlinear operation regime is shown in FIG. 1. In this setup the source and load impedances are nominally 50 Ω (the standard characteristic impedance Z_0 of microwave transmission lines). The input and output signal couplers (see ref. 3) extract a small portion of the RF forward and reverse power waves $a(t)$ and $b(t)$, which are injected into and extracted from the DUT, and transfers them into the tuned signal receiver (Vector Network Analyzer, VNA, see ref. 7), which measures the fundamental and harmonic components of it and may display the RF characteristics as well as the time function, using frequency-to-time inverse Fourier transformation. A tuner-coupler assembly allowing measuring the forward and reverse travelling waves $a(t)$ and $b(t)$ as a function of the reflection factor (RF impedance) presented to the DUT is called a “tuner for vector load pull” or “vector load pull tuner”.

(5) Directional signal couplers have been known for a long time (see ref. 1, 5). Waveguide directional couplers use sections of waveguide transmission lines and several slots or holes to allow small amounts of energy to leak into and excite wave generation and propagation inside a secondary adjacent waveguide (see ref. 5). The form, size and positioning of the slots or holes allows various coupling and directivity (that is the ratio between forward and reverse wave propagation) values. The main disadvantage of waveguide-to-waveguide couplers, as used in this invention, is their (typically) big size at low frequencies below 60 GHz, and because of that their main application is in millimeter and sub-THz frequency range between 110 and 330 GHz (0.11 to 0.33 THz). Directional signal couplers are, by principle, bi-directional, having a coupled and an isolated port; if the isolated port is not used, as is the case in scalar load pull and other applications, where the phase of the signal is not needed, then the isolated port is terminated with the characteristic impedance (Z_0) and ignored; the coupler becomes uni-directional. In this work the couplers are bi-directional.

BRIEF DESCRIPTION OF THE INVENTION

(6) This invention discloses a vector load pull tuner combining an integrated, compact and wideband, rectangular waveguide bi-directional signal coupler with a waveguide load-pull tuner. The signal coupler is made by attaching a secondary waveguide to a main waveguide, which is used to create the tuner, and extract a small portion (1% to 0.1%) of the signal power from the main waveguide into the secondary waveguide using holes of various number, forms, and disposition between the two waveguides allowing energy leakage. However, electro-magnetic waves do not propagate amorphaously, they follow the strict rules of Maxwell's wave propagation; the points across the communication holes excite wave propagation in the secondary waveguide and proper hole design, spacing and configuration allows obtaining adequate frequency coverage, coupling and directivity; in the remaining section, beyond the area of the slots or holes, of the main waveguide we create a load pull tuner, by cutting a longitudinal slot along the main waveguide and inserting a mobile conductive reflective probe inside this slot, to create a controlled reflection factor. The assembly allows vector load pull operation, because the bi-directional waveguide coupler allows measuring incident and reflected power waves into the DUT and thus calculating a large signal DUT impedance and delivered power and therefore the true power added efficiency (PAE), which a scalar load-pull system without a bi-directional coupler between tuner and DUT cannot do.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

- (1) The invention and its mode of operation will be more clearly understood from the following detailed description, when read with the appended drawings in which:
- (2) FIG. 1 depicts prior art, a signal (vector) measurement system using impedance tuners and bi-directional signal couplers.
- (3) FIG. 2 depicts prior art, the concept of the electro-magnetic wire sensor type bi-directional signal coupler in a waveguide.
- (4) FIG. 3 depicts prior art, a waveguide load pull tuner-wire coupler combination.
- (5) FIG. 4 depicts a waveguide load pull tuner-waveguide bi-directional coupler assembly.
- (6) FIG. 5 depicts the detailed view of the waveguide-slot based bi-directional coupler; the communication holes between the waveguides can have multiple forms and configurations.
- (7) FIG. 6 depicts coupling factor and directivity of the waveguide bi-directional coupler in the frequency range 110-170 GHz for various dimensions and configurations of the slot-shaped communication holes between the waveguides.
- (8) FIGS. 7A through 7C depict a number of evaluated communication holes: FIG. 7A depicts a rectangular slot arrangement; FIG. 7B depicts a round hole arrangement and FIG. 7C depicts a rectangular slot with two segments (one wide and one narrow) arrangement.
- (9) FIGS. 8A through 8B depict a further number of evaluated communication holes: FIG. 8A depicts a parallelogrammatical (rhombus) slot arrangement; FIG. 8B depicts a cross shaped slot arrangement.
- (10) FIGS. 9A through 9B depict two possible waveguide attachments and coupling mechanisms: FIG. 9A depicts a broad-to-narrow wall attachment and FIG. 9B a broad-to-broad wall attachment.
- (11) FIGS. 10A through 10B depict tuner-coupler (vector load pull tuner) assembly calibration setups: FIG. 10A depicts calibration using a 2-port vector network analyzer (VNA);
- (12) FIG. 10B depicts calibration using a 4-port VNA.
- (13) FIG. 11 depicts calibration flowchart of vector load pull tuner assembly using 2-port vector network analyzer.
- (14) FIG. 12 depicts calibration flowchart of vector load pull tuner assembly using 4-port vector network analyzer.

DETAILED DESCRIPTION OF THE INVENTION

- (15) Electro-magnetically coupled wire sensors for signal detection have been disclosed and used before (see ref. 5 and FIGS. 2 and 3). In prior art the electro-magnetic wire based signal couplers (terminating in ports 3, 4) are inserted in coaxial or waveguide 20 configuration (see ref. 4 and FIG. 2). However, there is a frequency limit to what can be accomplished using this technique, and that is the availability of microscopic size coaxial cables (diameter<0.8 mm), and adapters which are not available for frequencies above 140 GHz. For these higher frequencies the best (and only) solution are waveguide transmission lines and couplers. In the configuration of FIG. 3 a waveguide 31 tuner between ports 1 and 2 using a slot 33 and tuning probe 34 and an associated wire coupler with coupled ports 3 and 4: In this configuration the signal 30 exiting from the DUT is forward coupled 36 into the wire coupler and the reflected signal 32, 37 is reverse coupled into the same coupler before reaching the DUT to create the reflection factor T_{out} ; this way the incident 30 and reflected 37 power waves are assessed by the signal analyzer connected to ports 3 (forward) and 4 (reverse) of the coupler. As said, this system cannot exceed 140 GHz but the modus of operation is valid, if the associated coupler and tuner are available.
- (16) The overall waveguide-based tuner-coupler (vector load-pull tuner) assembly, operational up to at least 330 GHz, is shown in FIG. 4. The waveguide 43 has an input (or test) port 49, connected with the DUT and an output (or idle port) 40 connected with the load. In case of source pull the

same configuration applies, instead of the load port is connected to the source; the test port is always connected to the DUT. The bi-directional waveguide coupler having ports **401** and **402** is inserted between the test port **49** and the impedance tuner, represented here by the conductive tuning probe **42**, which is insertable vertically in-and-out of the slot **41** a to a multitude set of positions Y, and movable horizontally along the axis of the waveguide to a multitude of positions X, creating reflection factors covering a large portion of the reflection factor plan (Smith chart) at the test port. The tuning probe **42** creates a controlled reflection **44**, which sends energy back to the DUT. The bi-directional coupler is made using a secondary waveguide section **45**, which for the specific frequency range of 110-170 GHz uses the same standard rectangular size WR-6 (internal dimensions 0.065"×0.035" or 1.651×0.889 mm) as the main waveguide section **43**.

(17) The secondary waveguide **45** is separated from the main waveguide **43** by a thin common wall, which includes a number of communicating holes or openings **46** to allow the transfer of energy from the main to the secondary waveguide. The signal leaking through these holes excites electro-magnetic wave propagation into the secondary waveguide, the size and direction of which depends on the form of the openings and their mutual distance and configuration, as shown in FIGS. **7** and **8**; a small portion of the signal **48**, generated by the DUT, is coupled into the coupler **47**; most of the energy is fed into the forward coupled port **401** and a lower amount into the (reverse) isolated port **402**; the ratio of the two amounts is called "directivity" of the coupler and is a key characteristic of it. The higher the directivity the better the coupler; when the remaining signal **48** arrives at the tuning probe **42**, it is reflected **44** and returns to the DUT. In this flow direction, though, port **402** is the coupled port and port **401** the isolated port of the coupler. This way the measurement system can detect and determine the incident and reflected power waves from and into the DUT and calculate the effective reflection factor created by the tuner at the test port **49**, as well as the delivered power.

(18) FIG. **5** shows the coupling mechanism in detail: typical dimensions of the slots **53**, **54** and **72**, **73** for the WR-6 tuner covering 110-170 GHz are (FIG. **7A**): width **70** of 0.008" (0.2 mm) and length **71** of 0.025" (0.64 mm) or roughly 12% the width of the waveguide broad wall. If the waveguides are parallel but mounted vertical to each—other (FIG. **9A**) then the slot width is about 22% wide of the waveguide narrow wall. The main signal **52**, propagating through the main waveguide **55**, couples into the adjacent waveguide through the holes **53**, **54** . . . and exits through the coupled port **50** and the isolated port **51**. This allows detecting the propagating waves by an attached vector network analyzer and process into power and reflection factors. The waveguide sections can also be mounted either perpendicular to each other or under an angle between zero (parallel) and 90 degrees (perpendicular); several prior art publications have demonstrated the advantages or shortcomings of the various configurations.

(19) FIG. **6** shows coupling and directivity of a slot-based waveguide coupler per FIG. **7A** in a mounting configuration **90**, **91** per FIG. **9A** for various **60**, **61**, size combinations **70**, **71** and relative positionings **75** of the slots **72**, **73** machined into the common waveguide wall **74** of the main and secondary waveguides. It is obvious that the highest coupling factor $|S_{31}|$ (coupling from port 1 to port 3, FIG. **3**) and best frequency coverage in the 110-170 GHz range is associated with a directivity $|S_{31}|/|S_{32}|$ compromise ($|S_{32}|$ is reverse coupling between reverse input port 2 and forward coupled port 3). Nonetheless the directivity results of the waveguide coupler are, in general, better than those of the wire coupler, per FIG. **2**. FIGS. **9A** and **9B** show the two possible attachments of the two waveguides, the main waveguide **91**, **95**, which houses the tuner, and the secondary waveguide **90**, **94**, which houses the coupler, both communicating with various types and configurations of holes **92**, **93** or slots **80**, **81** (FIGS. **7A**, **B**, **C** and **8A**, **B**).

(20) A more detailed operation and definitions of a coupler-tuner assembly valid for wire loop based and for slot-based waveguide couplers is shown in FIG. **3**: the waveguide transmission line **31** has an input or test port 1 to which the DUT is connected, and an output port 2 connected to the load. The signal **30** flows from the DUT to the load. At the internal coupler port 1' a small portion

of the signal (typically 1% or less) is forward coupled **36** into the coupled port 3 of the signal coupler; the remaining signal power continues along the tuner section and is reflected **32** at the tuning probe **34**, which is inserted and manipulated inside the slot **33**. This reflected signal **37** reaches the wire coupler and leaks back **35** into the coupled port 3 and is added to the original forward coupled signal **36**; accurate measurement through the coupler requires accounting for this phenomenon. This leakage phenomenon leads to the following relation for the effective coupling factor $C_{31}'(\Gamma)$ between ports 1' and 3:

$$C_{31}'(\Gamma) = S_{31}' + S_{41}' * S_{21}' * \Gamma / (1 - \Gamma * S_{22}') \approx S_{31}' + S_{41}' * \Gamma \quad \{\text{eq.1}\}$$

(21) This simply means two things: a) that at $\Gamma=0$ the coupling is equal to the s-parameter S_{31}' and b) that for medium to low directivity $S_{31}'/S_{41}' = S_{31}'/S_{32}'$ the presence of Γ must be corrected for (see ref. 2).

(22) It would be possible to apply eq. 1 to correct C_{31}' , if S_{31}' and Γ were known. But in the integrated assembly, not only the real requirement is the external coupling factor C_{31} and not the internal C_{31}' , but also both the internal S_{31}' and Γ are unknown and cannot be measured. The low insertion loss of the waveguide might tempt one to estimate, with acceptable accuracy, the effective amplitude of $|S_{31}'|$ and $|\Gamma|$, but, since all signals dealt with here are vectors having amplitude and phase, absence of phase information is unacceptable. Therefore, a different method must be found yielding the much-needed information, i.e., an adequate system calibration.

(23) Calibration means prior characterization of a measurement instrument, or device, and saving the data in a way that can be recalled and referred to later. Impedance tuners, in general, are calibrated by measuring their two-port s-parameters from the input (test) port to the output (idle) port for a multitude of tuning probe positions, ideally the test port reflection factor S_{11} covering the whole or a large part of the reflection factor plan (Smith Chart), then save and recall the data (see ref. 6). The quantity of interest in the particular case of the couple-tuner assembly of FIG. 4 is, beyond the reflection factor S_{11} and tuner loss $(1 - |S_{11}|^2) / |S_{21}|^2$, the coupling and isolation factor C_{31} as a function of the reflection factor Γ , **44** as shown in eq. 1. As already explained, because the internal coupler ports 1', 2' (FIG. 3) and the value of Γ at its reference plane are inaccessible, a direct characterization of the internal coupler four-port (FIG. 5) is impossible. But this is irrelevant. What the measurement system needs to measure is the power “generated by” the DUT and the reflection factor “seen by” the DUT. This can be done at port 1 without further knowledge of the internal mechanism of the assembly. By measuring four-port s-parameters $S_{ij} = S_{11}, S_{12} \dots$ to S_{44} between the external ports 1 to 4 of the assembly as a function of the tuning probe positions, the whole system is characterized and calibrated. This line of thought is valid but not obvious, as a first reaction would be to characterize the coupler and tuner separately and compute the combination using theory as described by eq. 1. However, by scrutinizing the real requirement, we reach the conclusion that this, impossible, step is not even required.

(24) Tuner calibration requires a pre-calibrated vector network analyzer (VNA) (see ref. 7 and FIGS. **10A** and **10B**), connected using digital cables **100**, **104** and controlled by a system controller **101**, **103** which also controls **102**, **105** the tuner of the signal coupler tuner assembly. There are two basic types of VNA, 2-port VNA's having ports 1 and 2 and 4-port VNA's having ports 1 to 4; the difference is the internal duplication and multiplexing of signal measurement capability; when using a 4-port VNA (FIG. **12**) the calibration is simpler, because the four ports 1 to 4 of the coupler-tuner assembly are directly connected to the corresponding VNA ports and the 4-port s-parameters (between two ports at a time internally switched) are measured and saved in one operation as a function of a user selected multitude of tuning probe positions and saved in a calibration file A1234 containing 6 sets of two-port s-parameters per tuning probe position: $[A_{ij}(X,Y)]$ for $\{i,j\} = \{1,4\}$ and $i \neq j$, wherein $[A_{ij}]$ are matrices comprising four complex s-parameters each: $[A_{ij}] = [S_{11,ij}, S_{12,ij}, S_{21,ij}, S_{22,ij}]$. When using a simpler 2-port VNA (FIG. **11**), the calibration procedure becomes more tedious, because each time two of the four coupler-tuner assembly ports are connected to the VNA ports while the other two must be terminated with Z_0

(50Ω). In fact, s-parameters (see ref. 2) between the coupled and isolated ports 3 and 4 are not required for the operation, since these two ports are always Z_0 terminated and there is no signal power reflected to falsify the measurement. In summary a 4-port VNA requires connecting and measuring once, whereas a 2-port VNA (FIG. 11) requires connecting five, instead of six, times ($\{i-j\}=\{1-2, 1-3, 1-4, 2-3, 2-4\}$) two of the tuner-coupler assembly ports with ports 1 and 2 of the VNA, each time terminating the other two ports with Z_0 , measuring two-port s-parameters for the same set of tuning probe settings and saving in corresponding $[A_{ij}]$ matrices in associated calibration files A_{ij} . It is important that in this case the set of tuning probe positions is always the same. In the end the s-parameters of the A_{ij} files are concatenated to a total A_{1234} calibration file including 5 sets of four s-parameters to a total of 20 s-parameter sets. In the case of a 4-port VNA, s-parameter measuring and also saving s-parameters between ports 3 and 4 is a free bonus that does not cost considerable extra time. Most of the inconvenience when using 2-port VNA is the connecting disconnecting time and the 5-fold longer tuning probe positioning and measuring, in addition to unavoidable mechanical repeatability errors in positioning and re-positioning the tuning probe.

(25) Obvious alternatives of the disclosed embodiments of the slot-based waveguide coupler integrated with a waveguide tuner for sub-THz frequency vector load pull shall not impede on the reach of the invention. Obviously modified alternatives or re-arranged algorithms for calibration and for arranging the internal reference planes of the assembly shall not impede on the invention itself.

Claims

1. A waveguide vector load pull tuner comprising: a four-port assembly of a waveguide load pull tuner and a bi-directional waveguide signal coupler, wherein the waveguide load pull tuner comprises: a main rectangular waveguide having an input (test) port, an output (idle) port, a slot along a longitudinal axis of the main waveguide, and a remotely controlled conductive tuning probe, insertable perpendicularly into the slot between a state of withdrawal and a state of maximum penetration and movable inside the slot along the main waveguide over at least one half of a wavelength at a lowest frequency of operation of the waveguide vector load pull tuner, and wherein the bi-directional waveguide signal coupler comprises: a secondary rectangular waveguide terminating at a coupled and an isolated port touching the main waveguide at a shared wall region and communicating electro-magnetically with the main waveguide via a number of holes traversing the shared wall region of the main and the secondary waveguides, and wherein the slot of the main waveguide, in which the tuning probe penetrates, is placed between the idle port and the holes traversing the shared wall region and a calibration method of the waveguide vector load pull tuner using a two-port vector network analyzer (VNA); and a calibration method of the waveguide vector load pull tuner using a four-port vector network analyzer (VNA); said calibration methods generating and saving s-parameters of the vector load pull tuner four-port assembly as a function of frequency and a multitude of horizontal and vertical positions of the tuning probe.
2. The waveguide vector load pull tuner of claim 1, wherein the main and the secondary waveguides have a rectangular cross section with two broad walls and two narrow walls, and wherein a broad wall of the secondary waveguide is touching a broad wall of the main waveguide.
3. The waveguide vector load pull tuner of claim 1, wherein the main and the secondary waveguides have a rectangular cross section with two broad walls and two narrow walls, and wherein a narrow wall of the secondary waveguide is touching a broad wall of the main waveguide section.
4. The waveguide vector load pull tuner of claim 1, wherein the holes traversing the shared wall region of the main and the secondary waveguides are round.
5. The waveguide vector load pull tuner of claim 1, wherein the holes traversing the shared wall

region of the main and the secondary waveguides are rectangular slots.

6. The waveguide vector load pull tuner of claim 1, wherein the holes traversing the shared wall region of the main and the secondary waveguides are parallelogram slots.

7. The waveguide vector load pull tuner of claim 1, wherein the holes traversing the shared wall region of the main and the secondary waveguides are rectangular polygon slots having a wide section and a narrow section.

8. The waveguide vector load pull tuner of claim 1, wherein the holes traversing the common wall region of the main and the secondary waveguides are slots having the shape of a cross.

9. The calibration method for the waveguide vector load pull tuner as in claim 1, using a two-port vector network analyzer (VNA) having ports 1 and 2, pre-calibrated at a frequency F , comprising the following steps: a) the test and idle ports are connected to ports 1 and 2 of the VNA while the coupled and isolated ports are terminated with characteristic impedance (Z_0); b) two-port s-parameters S_{ij} for $\{i, j\}=\{1,2\}$ are measured at the frequency F for a multitude $M=N \times K$ of N horizontal (X) and K vertical (Y) positions of the tuning probe and saved in file A12 in the format $S_{ij}(X,Y)$; c) the test port and the coupled port are connected to ports 1 and 2 of the VNA while the idle port and the isolated port are terminated with characteristic impedance (Z_0); d) two-port s-parameters S_{ij} for $\{i, j\}=\{1,2\}$ are measured at the frequency F for the multitude $M=N \times K$ of the N horizontal (X) and the K vertical (Y) positions of the tuning probe and saved in file A13 in the format $S_{ij}(X,Y)$; e) the test port and the isolated port are connected to ports 1 and 2 of the VNA while the idle port and the coupled port are terminated with characteristic impedance (Z_0); f) two-port s-parameters S_{ij} for $\{i, j\}=\{1,2\}$ are measured at the frequency F for the multitude $M=N \times K$ of the N horizontal (X) and the K vertical (Y) positions of the tuning probe and saved in file A14 in the format $S_{ij}(X,Y)$; g) the idle port and the coupled port are connected to ports 1 and 2 of the VNA while the test port and the isolated port are terminated with characteristic impedance (Z_0); h) two-port s-parameters S_{ij} for $\{i, j\}=\{1,2\}$ are measured at the frequency F for the multitude $M=N \times K$ of the N horizontal (X) and the K vertical (Y) positions of the tuning probe and saved in file A23 in the format $S_{ij}(X,Y)$; i) the idle port and the isolated port are connected to ports 1 and 2 of the VNA while the test port and the coupled port are terminated with characteristic impedance (Z_0); j) two-port s-parameters S_{ij} for $\{i, j\}=\{1,2\}$ are measured at the frequency F for the multitude $M=N \times K$ of the N horizontal (X) and the K vertical (Y) positions of the tuning probe and saved in file A24 in the format $S_{ij}(X,Y)$; k) the coupled port and the isolated port are connected to ports 1 and 2 of the VNA while the test port and the idle port are terminated with characteristic impedance (Z_0); l) Two-port s-parameters S_{ij} for $\{i, j\}=\{1,2\}$ are measured at the frequency F for the multitude $M=N \times K$ of the N horizontal (X) and the K vertical (Y) positions of the tuning probe and saved in file A34 in the format $S_{ij}(X,Y)$; m) s-parameters $S_{ij}(X,Y)$ in files A12, A13, A14, A23, A24 and A34 are concatenated creating a calibration file A1234 of the vector load pull tuner.

10. The calibration method for the vector load pull tuner as in claim 1, using a four-port vector network analyzer (VNA) having ports 1, 2, 3 and 4, pre-calibrated at a frequency F , comprising the following steps: a) connect the test port to port 1, the idle port to port 2, the coupled port to port 3 and the isolated port to port 4; b) measure six sets $[A_{ab}]$, where $\{a,b\}=\{1,2,3,4\}$ and $a \neq b$, A12, A13, A14, A23, A24 and A34, of two-port s-parameters S_{ij} with $\{i,j\}=\{1,2\}$ at the frequency F and a multitude $M=N \times K$ of N horizontal (X) and K vertical (Y) positions of the tuning probe and save in a calibration file A1234 in a format $S_{ij}(X,Y)$.

11. The waveguide vector load pull tuner of claim 1, wherein the main waveguide and the secondary waveguide are mounted parallel to each other.

12. The waveguide vector load pull tuner of claim 1, wherein the main waveguide and the secondary waveguide are mounted perpendicular to each other.

13. The waveguide vector load pull tuner of claim 1, wherein the main waveguide and the secondary waveguide are mounted at an angle between zero and 90 degrees to each other.
